
Thermal camera Modbus/TCP communication protocol

1、Description

This device supports temperature measurement at 10 points and 10 areas.

The register address of Modbus TCP, with a valid value range (decimal) of [10191138].

One temperature measurement data occupies two register addresses.

The first register address stores the low 16 bits of data, and the second register address stores the high 16 bits of data.

The printing rule for register addresses from small to large is:

Point temperature measurement N+area temperature measurement N (minimum, maximum, average)+retention N

Temperature measurement register address printing table

No.	Point temperature measurement Register Address	Regional temperature measurement Register address			Reserved register address
		Min	Max	Avg	
1	1019--1020	1021--1022	1023--1024	1025--1026	1027--1030
2	1031--1032	1033--1034	1035--1036	1037--1038	1039--1042
3	1043--1044	'''	'''	'''	'''
4	1055--1056	'''	'''	'''	'''
5	1067--1068	'''	'''	'''	'''
6	1079--1080	'''	'''	'''	'''
7	1091--1092	'''	'''	'''	'''
8	1103--1104	'''	'''	'''	'''
9	1115--1116	'''	'''	'''	'''
10	1127--1128	1129--1130	1131--1132	1133--1134	1135--1138

2、Example

Through the webpage, set 1 point for temperature measurement and 1 area for temperature measurement. The actual measurement values are as follows:

Point temperature measurement 1: temperature 24.9 °

Regional temperature measurement 1: minimum temperature 22.5 ° , maximum temperature 29.7 ° , average temperature 25.0 °

Reminder:

The temperature measurement data uses the float type and is stored in the small end format. endianness conversion is not performed during network transmission.

Content	Value	Memory storage (small end format)
Point	24.9	33 33 C7 41
Area-Min	22.5	00 00 B4 41
Area-Max	29.7	9A 99 ED 41
Area-Avg	25.0	00 00 C8 41



Attention:

The register address of Modbus TCP starts from 1, and when the client sends a request, the address starts from 0,

Therefore, the value set on the program interface will decrease by 1 during actual transmission.

1019 (decimal)=0x3FB (hexadecimal)

1018 (decimal)=0x3FA (hexadecimal)

Holding Register	Word	Hex	Bin	Smallint	Float	Double	Smallint(Swap)	Float(Swap)	Double(Swap)
41019	13107	0x3333	0011 0011 0011 0011	13107	24.9	3.35544E8	13107	0.0	7.78463E8
41020	51009	0xC741	1100 0111 0100 0001	-14527			16839		
41021	0	0x0000	0000 0000 0000 0000	0	22.5		0	0.0	
41022	46145	0xB441	1011 0100 0100 0001	-19391			16820		
41023	39577	0x9A99	1001 1010 1001 1001	-25959	29.7	8.05306E8	-26214	0.0	3.97285E9
41024	60737	0xED41	1110 1101 0100 0001	-4799			16877		
41025	0	0x0000	0000 0000 0000 0000	0	25.0		0	0.0	
41026	51265	0xC841	1100 1000 0100 0001	-14271			16840		
41027	0	0x0000	0000 0000 0000 0000	0	0.0	0.0	0	0.0	0.0
41028	0	0x0000	0000 0000 0000 0000	0			0		
41029	0	0x0000	0000 0000 0000 0000	0	0.0		0	0.0	
41030	0	0x0000	0000 0000 0000 0000	0			0		

Modbus TCP client request command format

	Item	Protoco 1	Length	Device ID	Com mand	Register address	Register number
Occupancy length	2	2	2	1	1	2	2
Example	00 00	00 00	00 06	01	03	03 FA	00 0C

Modbus TCP device end reply format

	Item	Protoco 1	Length	Device ID	Comman d	Data length	Data
Occupancy length	2	2	2	1	1	1	N
Example	00 00	00 00	00 1B	01	03	18	----

Modbus client sends device commands:

00 00 00 00 00 06 01 03 03 FA 00 0C

Modbus client received device response:

00 00 00 00 **00 1B 01 03 18**

33 33 C7 41

00 00 B4 41 9A 99 ED 41 00 00 C8 41

00 00 00 00 00 00 00 00
